

JEE MAIN REVISION MASTER: WAVES SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 96 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Wave Equation & Velocity of Sound	62 PYQs	Wave Motion
Very High	Standing Waves, Resonance & Doppler Effect	46 PYQs	Interference & Acoustics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Wave Equation & Velocity of Sound	$y(x,t) = A \sin(kx \mp \omega t + \phi)$, $v = \frac{\omega}{k} = \lambda$, $v_{\text{string}} = \sqrt{\frac{T}{\mu}}$, $v_{\text{gas}} = \sqrt{\frac{\gamma RT}{M}}$
Standing Waves, Resonance & Doppler Effect	$f_{\text{open}} = \frac{nv}{2L}$; $(n=1,2,3)$, $f_{\text{closed}} = \frac{(2n-1)v}{4L}$; $(n=1,2,3)$, $f' = f \left(\frac{v \pm v_0}{v \mp v_s} \right)$, $f_{\text{beat}} = f_1 - f_2 $

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Wave Equation & Velocity of Sound (Very High Yield — 62 PYQs)

Core Principle: Progressive traveling harmonic waves, wave vector $k = 2\pi/\lambda$, and Laplace-corrected speed of sound in gases.

■ **Caution / JEE Pitfall:** Maximum particle velocity $v_{\text{(particle, max)}} = A\omega$ is completely different from wave propagation velocity $v = f\lambda$.

■ **Speed Trick:** Wave travels in positive x-direction if signs of kx and ωt are OPPOSITE: $\sin(kx - \omega t)$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Two strings (A,B) having linear densities $\mu_{\text{(A)}}=2 \times 10^{-4}\text{kg/m}$ and, $\mu_{\text{(B)}}=4 \times 10^{-4}\text{kg/m}$ and lengths $L_{\text{(A)}}=2.5\text{m}$ and ..." [Ans: B]
- [2025 Morning Shift] Q#1 (MCQ): "Two harmonic waves moving in the same direction superimpose to form a wave $x = a \cos(1.5t) \cos(50.5t)$ where t is in seconds. Find the period $w...$ " [Ans: N/A]
- [2025 Evening Shift] Q#1 (MCQ): "A sinusoidal wave of wavelength 7.5 cm travels a distance of 1.2 cm along the x-direction in 0.3 sec. The crest P is at $x=0$ at $t=0\text{sec}...$ " [Ans: N/A]

3.2 Standing Waves, Resonance & Doppler Effect (Very High Yield — 46 PYQs)

Core Principle: Stationary waves in strings and organ pipes, beat frequency, and Doppler frequency shift for moving sources and observers.

■ **Caution / JEE Pitfall:** A closed organ pipe produces ONLY odd harmonics (1, 3, 5...). The first overtone is the 3rd harmonic ($3v/4L$).

■ **Speed Trick:** Doppler rule: When distance decreases, apparent frequency INCREASES; when distance increases, apparent frequency DECREASES.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "The equation of a plane progressive wave is given by $y=5\cos\pi(200t-(x/150))$ where x and y are in cm and t is in second. The $v...$ " **[Ans: D]**
- **[2026 Evening Shift] Q#2 (MCQ):** "The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed p..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#3 (MCQ):** "A point source is kept at the center of a spherically enclosed detector. If the volume of the detector increased by 8 times, the intensity $w...$ " **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

The equation of a plane progressive wave is given by $y=5\cos\pi(200t-(x/150))$ where x and y are in cm and t is in second. The velocity of the wave is _____ m/s.

- (A) 120
- (B) 150
- (C) 200
- (D) 300 **[Correct]**

Correct Answer: D

Solution Analysis: A plane progressive wave moving in the positive x -direction can be represented as: $y=A\cos(\omega t-kx)$ Where, ω is the angular frequency, k is the propagation constant (wave number). The given equation is $y=5\cos\pi(200t-(x/150))$. $y=5\cos(200\pi t-(\pi x/150))$ By comparing this with $y=A\cos(\omega t-kx)$, $\omega=200\pi$ rad/s and $k=(\pi/150)$ rad/cm Wave velocity is the distance travelled by the wave per unit time. It is related to angular frequency and wave number by the relation: $v=(\omega/k)$ Substituting the values of ω and k : $v=\frac{200\pi}{\pi/150}$

Benchmark Problem 2: JEE Main 2026 (Online) 21st January Morning Shift (Q#1)

Two strings (A,B) having linear densities $\mu_A = 2 \times 10^{-4} \text{ kg/m}$ and $\mu_B = 4 \times 10^{-4} \text{ kg/m}$ and lengths $L_A = 2.5 \text{ m}$ and $L_B = 1.5 \text{ m}$ respectively are joined. Free ends of A and B are tied to two rigid supports C and D, respectively creating a tension of 500 N in the wire. Two identical pulses, sent from C and D ends, take time t_1 and t_2 , respectively, to reach the joint. The ratio t_1/t_2 is:

- (A) 1.90
- (B) 1.18 **[Correct]**
- (C) 1.08
- (D) 1.67

Correct Answer: B

Solution Analysis: When a string with linear mass density μ (mass per unit length) is under tension T , the speed v of a transverse wave is given by : $v = \sqrt{(T / \mu)}$ So, the time taken to reach the joint is $t = (\text{Length} / \text{Velocity}) = (L / v)$ Substituting the velocity formula into the time equation : $t = L \sqrt{(\mu / T)} = L \sqrt{(\mu / T)}$ The two strings are joined together and tied between two rigid supports, meaning the tension (T) is the same throughout both strings ($T = 500 \text{ N}$). For String A (time t_1) : The length of the string A is $L_A = 2.5 \text{ m}$ and the